

Muhammad Adeel Haider

Islamabad, Pakistan +92 3306869777 adeelhaider304@gmail.com github.com/BitR1ft linkedin.com/in/4deel

Professional Summary

Cybersecurity undergraduate with hands-on experience in penetration testing, detection engineering, and security automation through government VAPT work, independent research, and competitive security platforms. Performed authorized security assessments for the NCCIA, identifying critical web application vulnerabilities and producing remediation-focused reports. Built security tooling including a self-hosted SIEM, behavioral detection systems, and network analysis platforms. Active HackerOne researcher and CTF competitor with strong Python scripting skills and a growing focus on offensive security and threat detection.

Experience

VAPT Intern – *National Cyber Crime Investigation Agency NCCIA HQ, Islamabad*

Jun 2026 – Aug 2026

- Performed authorized penetration testing against government web applications, identifying four critical vulnerabilities aligned with OWASP Testing Guide methodologies.
- Expanded the known external attack surface by discovering previously undocumented subdomains using OSINT, directory enumeration, and service fingerprinting techniques.
- Successfully validated high-severity vulnerabilities, including SSRF and IDOR, despite deployed security controls, and documented reproducible proof-of-concept exploit paths.
- Delivered CVSS-based vulnerability reports with remediation guidance, enabling teams to reproduce and address issues.

Certifications & Achievements

- ISO/IEC 27001:2022 Lead Auditor** - MasterMind Certification
- Bug Bounty** - HackerOne (*BitR1ft*): Reported security findings across public bug bounty programs
- HackTheBox** - Completed all attack chain labs of Intro to Red Team and CPTS Preparation Tracks
- CTF Competitions** - Placed in BlackHat MEA 2025 qualifiers, AirTech, Softec, MetaCTF, Nexus, CSL
- Cryptography Research** - Designed and implemented an image encryption algorithm for academic research.

Technical Skills

SIEM & Detection Engineering: Splunk, Wazuh, Elastic Stack (ELK), Sigma Rules, MITRE ATT&CK

Threat Intelligence: MISP, OpenCTI, VirusTotal API, AbuseIPDB, IOC enrichment

Penetration Testing: Burp Suite, Nmap, ffuf, nuclei, sqlmap, Metasploit, Hydra, OpenVAS and more

Digital Forensics: Wireshark, PCAP analysis, TCP session reconstruction, evidence preservation

Programming & Scripting: Python (primary), C++, Bash

Key Projects

SentinelHive - SelfHosted SIEM & Threat Hunting Platform *Independent Research*

Jan 2026

- Built a self-hosted SIEM using Splunk, Wazuh, and the Elastic Stack to monitor authentication events, endpoint telemetry, and network activity across multiple endpoints.
- Developed over 40 Sigma detection rules mapped to MITRE ATT&CK and validated them against realistic attack scenarios generated in Hack The Box labs.
- Automated alert enrichment using MISP, OpenCTI, VirusTotal, and AbuseIPDB to improve investigation efficiency and reduce analyst workload.

SecureVPN - PostQuantum WireGuard VPN *Network Security Project*

May 2026

- Engineered a WireGuard VPN solution using a hybrid post-quantum key exchange combining Kyber-512 and X25519.
- Built a Python Flask provisioning service supporting secure peer management, replay protection, Redis-based rate limiting, and Docker deployment.
- Developed a Windows client featuring automatic configuration, secure key storage using DPAPI, DNS leak protection, and a configurable kill switch.

SentinelTrace - Linux Syscall Tracing & Behavioral Analysis *Operating Systems Project*

April 2026

- Developed a Linux behavioral monitoring system using ptrace to collect and analyze process syscall activity.
- Implemented a YAML-driven detection engine capable to identify suspicious syscall sequences and behavioral anomalies.
- Built a PyQt6 dashboard with SQLite-backed forensic replay to visualize alerts and investigate process behavior.

NetSpector - Network Traffic Analyzer *Computer Networks Project*

Dec 2025

- Developed a packet analysis platform for PCAP and PCAPNG inspection for incident response and network forensics.
- Implemented detection rules for SYN floods, port scans, DNS tunneling, credential leakage, and anomalous traffic patterns, enhancing proactive threat identification and strengthening incident response workflows.
- Added TCP session reconstruction, bidirectional flow analysis, and file carving capabilities.

Education

Bachelor of Science in Cyber Security (BSCYS) - *Air University, NCSA, Islamabad*

Aug 2024 – Present